Question	Answer	Marks	Guidance
6(a)	• force per unit length • force per unit current • length / current perpendicular to field *1 mark for any two bullet points, 2 marks for all three bullet points.*	B2	Ratio must be clear. **Ignore** symbols unless defined. **Ignore** references to units. 'Force per unit (current × length)' achieves both BP1 and BP2. **Allow** 'wire perpendicular to field'. **Allow** 'flux (density)' for 'field'. **Do not allow** 'force' for 'field'.
6(b)(i)	(at X, magnetic) force on particles is to the left	B1	
	left hand rule tells us the (effective) current is in the direction of movement (of the particles), so charge is positive	B1	**Allow** 'beam' or 'velocity' for 'movement' **Allow** 'down' for 'direction of movement'
6(b)(ii)	(magnetic) force is always perpendicular to the direction of motion (of the particles)	B1	**Allow** 'beam' or 'velocity' for 'direction of motion'. **Allow** 'centripetal' for 'magnetic'
	(magnetic) force changes direction only, and not the speed of the particles **or** (magnetic) force causes centripetal acceleration	B1	**Allow** 'centripetal' for 'magnetic'
6(c)(i)	$a = v^2 / r$	C1	**Allow** $F = ma$ **and** $F = mv^2 / r$
	$= (7.3 \times 10^4)^2 / (0.5 \times 8.2 \times 10^{-2}) = 1.3 \times 10^{11}\ \text{m s}^{-2}$	A1	Full substitution and answer needed. Allow 0.041 for $0.5 \times 8.2 \times 10^{-2}$ Unit not needed, but if a unit is given it must be correct.

Question	Answer	Marks	Guidance
6(c)(ii)	$BQv = ma$	C1	**Allow** $Bq = mv / r$. No ECF from **6(c)(i)**.
	$B = (2.0 \times 10^{-26} \times 1.3 \times 10^{11}) / (3.2 \times 10^{-19} \times 7.3 \times 10^{4})$ = 0.11 T	A1	Correct to at least 2 significant figures. AFC. (0.111) Unit needed with answer for A1 mark. **Allow** kg A^{-1} s^{-2} for T **Allow** kg C^{-1} s^{-1} for T **Allow** N A^{-1} m^{-1} for T **Allow** Wb m^{-2} for T **Allow** V s m^{-2} for T
6(d)	beam shown entering the field at X along same line, and deflecting in same direction in field	B1	
	circular path with half the radius along the whole length of the curve	B1	Judge half the radius approximately by eye